Mark Hedley Jones

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Osaka, Japan (permanent resident); New Zealand and British citizen

Education

2010–2016 **PhD in Engineering**, University of Waikato, New Zealand.
Thesis: *The Electrical Properties of Interfacial Double Layers*. Doctoral Scholarship. Part-time from 2015 alongside post-doctoral work; thesis submitted August 2016, conferred December 2016.

2009–2010 **Master of Engineering, First Class Honours**, University of Waikato.
Modelled a millimetre-wave power sensor and tested the simulations against a physical scale model. Agilent Technologies research scholarship.

2005–2008 **Bachelor of Science, Electronics**, University of Waikato. Dean's Honours List, 2008.

Software

Tools Python and C++ (over ten years each), NumPy, SciPy, Matplotlib, pytest, Git, GitHub Actions, Linux, Docker, PyTorch, OpenCV.

2026 **Cartographer trajectory editor**: browser tool for correcting robot maps. Implements Levenberg–Marquardt least-squares optimisation in JavaScript.

2015–2016 **Electrode Interface Model**: Python model of electrode impedance, using ngSpice circuit simulations and SciPy optimisation to fit measurements from saline and living tissue.

2025–present **container-magic**: Python tools for reproducible containerised development environments; automated tests with pytest and GitHub Actions.

2013–present **dmenu-extended**: Python application and file launcher for Linux, with automated tests and continuous integration.

2013 **pyLabInstruments**: Python library for automating laboratory instruments and measurements.

2018–present **Calibration checkerboard collection**: printable camera-calibration targets and a generator, used internationally in robotics.

Employment

2018–present **Software Engineer**, SEQSENSE, Tokyo, Japan.
Software for autonomous security and delivery robots.

- Lidar-based simultaneous localisation and mapping (SLAM) software for desktop, cloud and on-robot use.
- 3D object detection and tracking from lidar and camera data; models trained in PyTorch and deployed on robots.
- Gazebo robot simulations built from mechanical designs.

2018 **Research Fellow**, University of Auckland. January–September 2018.
Developed navigation software for kiwifruit orchards, where canopies block satellite positioning: lidar ground segmentation, row detection and turn planning using Nelder–Mead optimisation. Compared Cartographer settings through scripted parameter sweeps over recorded sensor data.
Write-up: markhedleyjones.com/projects/autonomous-navigation-kiwifruit-orchards

2015–2018 **Post-Doctoral Researcher**, University of Waikato. April 2015–January 2018.
Developed a research vehicle with a 1000 kg payload capacity for kiwifruit harvesting and pollination. Took it from an early chassis design to field trials, working with university and industry colleagues on fabrication, electrical assembly and vehicle-control software.
Write-up: markhedleyjones.com/projects/robotics-platform-kiwifruit-orchards

2009–2014 **Sessional Assistant**, University of Waikato. Electronics laboratory demonstrator; developed an online tutorial website for first-year electronic engineering students. Write-up: markhedleyjones.com/projects/electronics-tutorial-website

Publications

Also at markhedleyjones.com/publications | ORCID 0000-0003-3387-2932

Journal articles

1. **Jones, M. H.**, & Scott, J. (2015). Scaling of electrode–electrolyte interface model parameters in phosphate buffered saline. *IEEE Transactions on Biomedical Circuits and Systems*, 9(3), 441–448. doi:10.1109/TBCAS.2014.2333759
2. **Jones, M. H.**, Bell, J., Dredge, D., Seabright, M., Scarfe, A., Duke, M., & MacDonald, B. (2019). Design and testing of a heavy-duty platform for autonomous navigation in kiwifruit orchards. *Biosystems Engineering*, 187, 129–146. doi:10.1016/j.biosystemseng.2019.08.019
3. Williams, H. A. M., **Jones, M. H.**, et al. (2019). Robotic kiwifruit harvesting using machine vision, convolutional neural networks, and robotic arms. *Biosystems Engineering*, 181, 140–156. doi:10.1016/j.biosystemseng.2019.03.007
4. Williams, H., Ting, C., Nejati, M., **Jones, M. H.**, et al. (2020). Improvements to and large-scale evaluation of a robotic kiwifruit harvester. *Journal of Field Robotics*, 37(2), 187–201. doi:10.1002/rob.21890
5. Williams, H., Nejati, M., Hussein, S., Penhall, N., Lim, J. Y., **Jones, M. H.**, et al. (2020). Autonomous pollination of individual kiwifruit flowers: Toward a robotic kiwifruit pollinator. *Journal of Field Robotics*, 37(2), 246–262. doi:10.1002/rob.21861

Conference papers, posters and book chapter

6. **Jones, M. H.**, Seabright, M., Barnett, J., Neshausen, G., Duke, M., & Scarfe, A. (2017). An electrically driven, computer controlled robotics platform for orchard use. Poster, International Tri-Conference for Precision Agriculture. doi:10.5281/zenodo.1002306
7. Seabright, M., **Jones, M. H.**, et al. (2017). Robotic harvesting of kiwifruit. Poster, International Tri-Conference for Precision Agriculture. doi:10.5281/zenodo.1002294
8. Hicks, B., **Jones, M. H.**, de Villiers, J., & Ling, N. (2015). Use of electrofishing for capturing invasive fish. In *New Zealand Invasive Fish Management Handbook*. Lake Ecosystem Restoration New Zealand and Department of Conservation.
9. **Jones, M. H.**, & Scott, J. (2014). Feasibility of harvesting power to run a domestic water meter using streaming cell technology. *21st Electronics New Zealand Conference*, Hamilton. Runner-up, best presentation.
10. **Jones, M. H.**, & Scott, J. (2011). The energy efficiency of 8-bit low-power microcontrollers. *18th Electronics New Zealand Conference*, Palmerston North. Best research paper.
11. **Jones, M. H.**, & Scott, J. (2011). Design study of a thermocouple power sensor as a monolithic fin-line. *77th ARFTG Microwave Measurement Conference*, Baltimore. doi:10.1109/ARFTG77.2011.6034554